

GECON 7201

Applied Econometrics

Assignment 1
Due Thursday October 8, 2026

Trinh Nhat Minh Le

Instructions

- Edit this Quarto document for your assignment. All answers must be typed in this document. Answers are to be written in the callout blocks (see below)
- Provide a link to your website in Nexus, so that I can view your assignment.
- Upload both a PDF and Quarto file to the nexus assignment portal.
- The use of AI is strictly prohibited for this assignment.
- Show your work
- Type your mathematical answers using LaTeX
- For questions marked R, include both your code and its output.
- You may use `solve()`, `t()`, `%*%`, `diag()`, `qr()`, `eigen()`, and other base R functions freely unless a question says otherwise.
- Please round numerical answers to two decimal places unless stated otherwise.

```
set.seed(3201)
```

Questions

Q1

Create a personal Quarto website hosted on GitHub pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

- a. An “Index” or “Home” page. This page should include the following: - Your name - Your photo (optional) - Your credentials - An “About Me” section that provides a brief (one paragraph) introduction to yourself. [10 Marks]

- b. A blog page. This will be the landing site for future assignments [5 Marks]
- c. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
- d. Navigation bar with “Home” and “Blog” buttons. The navigation bar is edited in the “_quarto.yml” file. [2 Marks]

<https://minhclone4641.github.io/Minh-GECON-7201/>

Q2

Consider

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -3 \\ 0 & -3 & 6 \end{bmatrix}.$$

- a. State the dimensions of A and identify a_{23} . [2 Marks]

The matrix A has 3 rows and 3 columns. Therefore:

$$\dim(A) = 3 \times 3.$$

The entry a_{23} is in row 2 and column 3:

$$a_{23} = -1.$$

- b. Is A a square matrix? Is it symmetric? Justify your answer using the definition $A = A^\top$, not just by inspection. [2 Marks]

Your answer here

A is square because it has the same number of rows and columns.

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

Taking the transpose switches the rows and columns:

$$A^\top = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} = A.$$

$\Rightarrow A = A^\top$, so A is symmetric.

- c. Is A a diagonal matrix? Explain why or why not. [1 Mark]

A is not diagonal because some entry outside the main diagonal are not zero. For example, $a_{13} = 1$ and $a_{23} = -1$.

- d. In R, Enter A and B into R. Use `isSymmetric()` to confirm your answer to part (b) for both matrices. [4 Marks]

```
#Enter matrices A and B
A <- matrix(c(2, 0, 1,
              0, 3, -1,
              1, -1, 4),
            nrow = 3)
```

```
B <- matrix(c(1, 2, 0,
              2, 5, -3,
              0, -3, 6),
            nrow = 3)
```

```
#Print matrices
```

A

	[,1]	[,2]	[,3]
[1,]	2	0	1
[2,]	0	3	-1
[3,]	1	-1	4

B

	[,1]	[,2]	[,3]
[1,]	1	2	0
[2,]	2	5	-3
[3,]	0	-3	6

```
#Check symmetry
```

```
isSymmetric(A)
```

```
[1] TRUE
```

```
isSymmetric(B)
```

```
[1] TRUE
```

Results show TRUE $\Rightarrow A$ and B are symmetric.

Q3

Using the same A and B from Question 1:

- a. Compute $A + B$ and $3A - 2B$ by hand. [4 marks]

$$A + B = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 8 & -4 \\ 1 & -4 & 10 \end{bmatrix}.$$

Also:

$$3A - 2B = \begin{bmatrix} 4 & -4 & 3 \\ -4 & -1 & 3 \\ 3 & 3 & 0 \end{bmatrix}.$$

- b. Compute AB by hand, showing at least one entry's calculation in full using the summation formula $(AB)_{ij} = \sum_{\ell} a_{i\ell} b_{\ell j}$. [2 marks]

To begin with, the entry in row 1 and column 2 is:

$$(AB)_{12} = 2(2) + 0(5) + 1(-3) = 1.$$

Calculating the remaining entries in the same way:

$$AB = \begin{bmatrix} 2 & 1 & 6 \\ 6 & 18 & -15 \\ -1 & -15 & 27 \end{bmatrix}.$$

- c. Compute BA . Is $AB = BA$ here? Was this guaranteed in general? [2 marks]

$$BA = \begin{bmatrix} 2 & 6 & -1 \\ 1 & 18 & -15 \\ 6 & -15 & 27 \end{bmatrix}.$$

Therefore:

$$AB \neq BA.$$

$\Rightarrow$ Matrix multiplication is not commutative in general.

- d. In R, verify parts (a)–(c) in R. Also compute $A * B$ (element-wise) and explain in one sentence why this is a different operation from $A \%*\% B$. [4 marks]

```
#part a sum
A_plus_B <- A + B
A_plus_B
```

```
      [,1] [,2] [,3]
[1,]    3    2    1
[2,]    2    8   -4
[3,]    1   -4   10
```

```
#part a: 3A - 2B
three_A_minus_two_B <- 3*A - 2*B
three_A_minus_two_B
```

```
      [,1] [,2] [,3]
[1,]    4   -4    3
[2,]   -4   -1    3
[3,]    3    3    0
```

```
#part b and c
AB <- A %*% B
BA <- B %*% A
AB
```

```
      [,1] [,2] [,3]
[1,]    2    1    6
[2,]    6   18  -15
[3,]   -1  -15   27
```

```
BA
```

```
      [,1] [,2] [,3]
[1,]    2    6   -1
[2,]    1   18  -15
[3,]    6  -15   27
```

```
#part c
all.equal(AB, BA)
```

```
[1] "Mean relative difference: 1.714286"
```

```
print("=> AB is not equal to BA")
```

```
[1] "=> AB is not equal to BA"
```

```
#Element-wise multiplication
```

```
A_times_B <- A * B
```

```
A_times_B
```

```
      [,1] [,2] [,3]  
[1,]     2     0     0  
[2,]     0    15     3  
[3,]     0     3    24
```

→ $A * B$ multiplies corresponding entries, while $A \%*\% B$ performs row-by-column matrix multiplication.

Q4

Let

$$X = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

a. Write down $X^\top$ and state its dimensions. [2 marks]

$$X = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \Rightarrow X^\top = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

The dimension of $X^\top$ is 2×3 .

b. Compute $X^\top X$ and $XX^\top$ by hand. Explain why these two products have different dimensions even though both involve X and $X^\top$. [4 marks]

$$X^\top X = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}.$$

$$XX^\top = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}.$$

$X^\top X$ has dimension $(2 \times 3)(3 \times 2) = 2 \times 2$, while $XX^\top$ has dimension $(3 \times 2)(2 \times 3) = 3 \times 3$.

- c. Let $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{v}^\top \mathbf{v}$ and $\mathbf{v}\mathbf{v}^\top$ by hand, and state the dimension of each result.
[4 marks]

Let:

$$v = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}.$$

Then:

$$v^\top = [1 \quad -1 \quad 2].$$

$$v^\top v = 1^2 + (-1)^2 + 2^2 = 6.$$

This is a scalar. Also:

$$vv^\top = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix}.$$

This result has dimension 3×3 .

- d. In R, verify parts (a)–(c). Then confirm numerically, for two conformable matrices of your choosing, that $(AB)^\top = B^\top A^\top$. [4 marks]

```
#Verify parts (a) - (c)
#Enter matrix X and vector v
X <- matrix(c(1, 4,
              2, 5,
              3, 6),
            nrow = 3)

v <- matrix(c(1, -1, 2), ncol = 1)

#Transpose and products
t(X)
```

```
      [,1] [,2] [,3]
[1,]    1    4    2
[2,]    5    3    6
```

```
t(X) %*% X
```

```
      [,1] [,2]  
[1,]    21    29  
[2,]    29    70
```

```
X %*% t(X)
```

```
      [,1] [,2] [,3]  
[1,]    26    19    32  
[2,]    19    25    26  
[3,]    32    26    40
```

```
t(v) %*% v
```

```
      [,1]  
[1,]     6
```

```
v %*% t(v)
```

```
      [,1] [,2] [,3]  
[1,]     1    -1     2  
[2,]    -1     1    -2  
[3,]     2    -2     4
```

⇒ matches with the results above.

```
#Choose conformable matrices
```

```
A <- matrix(c(1, 2, 3,  
              4, 5, 6),  
            nrow = 2)
```

```
B <- matrix(c(1, 0,  
              2, 1,  
              3, -1),  
            nrow = 3)
```

```
#Verify transpose rule
```

```
left_side <- t(A %*% B)
```

```
right_side <- t(B) %*% t(A)
```

```
left_side
```

```

      [,1] [,2]
[1,]   11  14
[2,]    5   8

```

```
right_side
```

```

      [,1] [,2]
[1,]   11  14
[2,]    5   8

```

```
all.equal(left_side, right_side)
```

```
[1] TRUE
```

⇒ The result is **TRUE**, confirming $(AB)^T = B^T A^T$.

Q5

Consider

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}.$$

a. In R, compute A^{-1} using `solve()`. [2 marks]

```

#Enter matrix A
A_q5 <- matrix(c(4, 2,
                 3, 3),
              nrow = 2)

#Compute inverse
A_inverse <- solve(A_q5)
round(A_inverse, 2)

```

```

      [,1] [,2]
[1,]  0.50 -0.50
[2,] -0.33  0.67

```

Therefore:

$$A^{-1} = \begin{bmatrix} 0.50 & -0.33 \\ -0.50 & 0.67 \end{bmatrix}.$$

- b. Construct a 3×3 matrix that is singular (for example, make one column a linear combination of the others) and show what happens when you call `solve()` on it. [2 marks]

```
#Construct a singular matrix
B_q5 <- matrix(c(1, 2, 3,
                 2, 4, 6,
                 3, 6, 9),
              nrow = 3)

B_q5
```

```
      [,1] [,2] [,3]
[1,]     1     2     3
[2,]     2     4     6
[3,]     3     6     9
```

```
#Try to compute its inverse
solve(B_q5)
```

Error in `solve.default(B_q5)`: Lapack routine `dgesv`: system is exactly singular: U[2,2] = 0

The rows are linear combinations of the others. Therefore, the matrix is singular and its inverse does not exist => R code shows error.

Q6

- a. Define, in your own words, what it means for a set of vectors to be linearly independent. [1 mark]

A set of vectors is linearly independent if the equation

$$c_1x_1 + c_2x_2 + \dots + c_kx_k = 0$$

has only the solution $c_1 = c_2 = \dots = c_k = 0$.

b. Consider

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}.$$

Show that $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly dependent by finding constants c_1, c_2, c_3 , not all zero, such that $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = \mathbf{0}$. [3 marks]

We have:

$$2x_1 + x_2 = 2 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix} = x_3.$$

Therefore:

$$2x_1 + x_2 - x_3 = 0.$$

One nonzero solution is $c_1 = 2$, $c_2 = 1$, and $c_3 = -1$.

$\Rightarrow$ The vectors are linearly dependent.

c. Let $M = [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3]$. What is $\text{rank}(M)$? Explain your reasoning using your answer to part (b). [2 marks]

Since $x_3 = 2x_1 + x_2$, the third column does not add a new independent direction. The first two columns are not multiples of each other.

$$\text{rank}(M) = 2.$$

d. In R, build M and confirm its rank using `qr(M)$rank`. Then modify one entry of $\mathbf{x}_3$ so that the three vectors become linearly independent, and confirm the new rank. [3 marks]

```
#Define vectors
x1 <- c(1, 2, 1)
x2 <- c(0, 1, 1)
x3 <- c(2, 5, 3)

#Build matrix M
M <- cbind(x1, x2, x3)
M

      x1 x2 x3
[1,]  1  0  2
[2,]  2  1  5
```

```
[3,] 1 1 3
```

```
#Confirm rank
```

```
qr(M)$rank
```

```
[1] 2
```

```
#Modify one entry of x3
```

```
x3_new <- c(2, 5, 4)
```

```
M_new <- cbind(x1, x2, x3_new)
```

```
M_new
```

```
      x1 x2 x3_new
[1,]  1  0      2
[2,]  2  1      5
[3,]  1  1      4
```

```
#Confirm new rank
```

```
qr(M_new)$rank
```

```
[1] 3
```

⇒ The original rank is 2. After changing the third entry of x_3 from 3 to 4, the rank becomes 3 ⇒ the new vectors are linearly independent.

Q7.

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

a. Write out the quadratic form $\mathbf{x}^\top A \mathbf{x}$ in terms of x_1 and x_2 . [2 marks]

$$\mathbf{x}^\top A \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

First:

$$A\mathbf{x} = \begin{bmatrix} 2x_1 + x_2 \\ x_1 + 2x_2 \end{bmatrix}.$$

Therefore:

$$x^\top Ax = 2x_1^2 + 2x_1x_2 + 2x_2^2.$$

- b. Is B invertible? Justify your answer using the relationship between definiteness, rank, and invertibility discussed in class. [3 marks]

For

$$B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

Let

$$b = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

$\Rightarrow$

$$B.b = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1-1 \\ 1-1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow B.b = 0.b$$

$\Rightarrow$ an eigenvalue is zero $\Rightarrow B$ does not have full rank:

$$\text{rank}(B) = 1 < 2.$$

A square matrix is invertible only if it has full rank. Therefore, B is not invertible.

- c. In R, Confirm parts (a) and (b) by computing `eigen(A)$values` and `eigen(B)$values`. State the rule connecting the sign of a symmetric matrix's eigenvalues to whether it is positive definite, positive semidefinite, or neither. [4 marks]

```
#Enter matrices
A_q7 <- matrix(c(2, 1,
                 1, 2),
               nrow = 2)

B_q7 <- matrix(c(1, 1,
                 1, 1),
               nrow = 2)

#Compute eigenvalues
eigen(A_q7)$values
```

```
[1] 3 1
```

```
eigen(B_q7)$values
```

```
[1] 2 0
```

```
#Confirm rank of B
```

```
qr(B_q7)$rank
```

```
[1] 1
```

The eigenvalues of A are 3 and 1, so A is positive definite. The eigenvalues of B are 2 and 0, so B is positive semidefinite and singular.

Q8

Let

$$A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

- a. Show by direct calculation that $A^2 = A$, i.e., that A is idempotent. [2 marks]

$$A^2 = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} = A.$$

Therefore, A is idempotent.

- b. Using only the idempotent property $A^2 = A$ (not this specific matrix), show that $I - A$ is also idempotent. [2 marks]

Using $A^2 = A$:

$$(I - A)^2 = I - 2A + A^2.$$

Substitute $A^2 = A$:

$$(I - A)^2 = I - 2A + A = I - A.$$

$\Rightarrow I - A$ is also idempotent.

- c. In R, confirm that A is idempotent using matrix multiplication in R. [1 mark] ::: callout-tip

```
#Enter matrix A
A_q8 <- matrix(c(0.5, 0.5,
                 0.5, 0.5),
               nrow = 2)

#Compute A squared
A_squared <- A_q8 %*% A_q8
A_squared
```

```
      [,1] [,2]
[1,]  0.5  0.5
[2,]  0.5  0.5
```

```
#Confirm idempotency
all.equal(A_squared, A_q8)
```

```
[1] TRUE
```

```
:::
```

Q9

Let $\mathbf{a} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and

$$S = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Note that S is symmetric.

- a. Write down $\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}}$ for a general 3×1 vector $\mathbf{x}$. (No calculation needed, state the rule and apply it to this $\mathbf{a}$.) [2 marks]

For a constant vector a :

$$\frac{\partial(a^\top x)}{\partial x} = a.$$

Therefore:

$$\frac{\partial(a^\top x)}{\partial x} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}.$$

- b. Derive $\frac{\partial(\mathbf{x}^\top S \mathbf{x})}{\partial \mathbf{x}}$ using the general rule for $\frac{\partial(\mathbf{x}^\top A \mathbf{x})}{\partial \mathbf{x}}$, and simplify using the fact that S is symmetric. [2 marks]

The general rule is:

$$\frac{\partial(x^\top A x)}{\partial x} = (A + A^\top)x.$$

Since S is symmetric $\Rightarrow S = S^\top$:

$$\frac{\partial(x^\top S x)}{\partial x} = (S + S^\top)x = (S + S)x = 2Sx.$$

- c. Evaluate your answer to part (b) at $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$. [1 mark]

At $x = (1, 0, -1)^\top$:

$$2Sx = 2 \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}.$$

- d. In R, using the `data:image/png;base64,iVBORw0KGgoAAAANSUhEUgAAABIAAAASCAYAAABWzo5XAAAAAbE1EQ` package (`grad()`), numerically confirm your answers to parts (a) and (c). [2 marks]

```
#Load package
library(numDeriv)
```

Warning: package 'numDeriv' was built under R version 4.5.2

```
#Define vector a and matrix S
a <- c(3, -2, 1)
```

```
S <- matrix(c(2, 1, 0,
              1, 3, 1,
              0, 1, 2),
            nrow = 3)
```

```
#Part (a): f(x) = a'x
f_a <- function(x) {sum(a*x)}
```

```
#Numerically confirm part (a)
grad(f_a, x = c(1, 2, 3))
```

```
[1] 3 -2 1
```

```
#Part (c): f(x) = x'Sx
f_c <- function(x) {
  as.numeric(t(x) %*% S %*% x)
}
```

```
#Numerically confirm part (c)
round(grad(f_c, x = c(1, 0, -1)),2)
```

```
[1] 4 0 -4
```

⇒ The numerical gradients are approximately $(3, -2, 1)^\top$ and $(4, 0, -4)^\top$, matching the manual results.

Q10

Let $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$ be a random vector with

$$\mu = E[\mathbf{X}] = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \quad \Sigma = \text{Var}(\mathbf{X}) = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}.$$

Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$, and define $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$.

a. State $E[\mathbf{Y}]$ and $\text{Var}(\mathbf{Y})$, citing the properties used. [3 marks]

For $Y = AX + b$:

$$E[Y] = AE[X] + b = A\mu + b,$$

and:

$$\text{Var}(Y) = A \text{Var}(X) A^\top = A\Sigma A^\top.$$

The constant vector b changes the mean but does not affect the variance.

b. Compute $E[\mathbf{Y}]$ by hand. [2 marks]

$$E[Y] = A\mu + b = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 5 \\ -2 \end{bmatrix}.$$

Therefore:

$$E[Y] = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 5 \\ -2 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}.$$

c. Compute $\text{Var}(\mathbf{Y}) = A\Sigma A^\top$ by hand, showing your intermediate matrix products. [3 marks]

First:

$$A\Sigma = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 3 & -1 \\ 1 & 3 & -4 \end{bmatrix}$$

Therefore:

$$A\Sigma A^\top = \begin{bmatrix} 5 & 3 & -1 \\ 1 & 3 & -4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 8 & 4 \\ 4 & 7 \end{bmatrix}.$$

Thus:

$$\text{Var}(Y) = \begin{bmatrix} 8 & 4 \\ 4 & 7 \end{bmatrix}.$$

- d. Let $\mathbf{a} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Compute $\text{Var}(\mathbf{a}^\top \mathbf{X})$ using the appropriate property, without recomputing $\text{Var}(\mathbf{X})$ from scratch. [3 marks]

Using $\text{Var}(a^\top X) = a^\top \Sigma a$:

$$\Sigma a = \begin{bmatrix} 3 \\ -3 \\ 7 \end{bmatrix}.$$

Therefore:

$$\text{Var}(a^\top X) = \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -3 \\ 7 \end{bmatrix} = 3 + 3 + 14 = 20.$$

- e. In R, using `MASS::mvrnorm()`, draw 100,000 observations from a distribution with mean μ and variance Σ (you may treat $\mathbf{X}$ as multivariate normal for this simulation). Confirm your answers to (b), (c), and (d) using `colMeans()` and `cov()`. [3 marks]

```

#Set seed
set.seed(3201)

#Define parameters
mu <- c(2, 0, -1)

Sigma <- matrix(c(4, 1, 0,
                  1, 2, -1,
                  0, -1, 3),
                nrow = 3, byrow = TRUE)

A_q10 <- matrix(c(1, 1, 0,
                  0, 1, -1),
                nrow = 2, byrow = TRUE)

b <- c(5, -2)
a_q10 <- c(1, -1, 2)

#Draw observations
X_sim <- MASS::mvrnorm(n = 10000,
                      mu = mu,
                      Sigma = Sigma)

#Construct Y = AX + b
Y_sim <- X_sim %*% t(A_q10) + matrix(b, nrow(X_sim), 2, byrow = TRUE)

#Simulated mean and variance
round(colMeans(Y_sim), 2)

[1] 6.99 -1.01

round(cov(Y_sim), 2)

      [,1] [,2]
[1,] 8.01 3.82
[2,] 3.82 6.91

#Construct a'X and calculate variance
ax_sim <- X_sim %*% a_q10
round(var(as.numeric(ax_sim)), 0)

```

[1] 20

The simulated results should be close to:

$$E[Y] = \begin{bmatrix} 7 \\ -1 \end{bmatrix}, \quad \text{Var}(Y) = \begin{bmatrix} 8 & 4 \\ 4 & 7 \end{bmatrix}, \quad \text{Var}(a^\top X) = 20.$$

Q11

- a. Suppose $\mathbf{X} \sim N_3(\mu, \Sigma)$ using the μ and Σ from Question 10. What is the marginal distribution of X_2 alone? Give its distribution and parameters. [2 marks]

A component of a multivariate normal vector is normally distributed. From μ and Σ :

$$E[X_2] = \mu_2 = 0$$

and:

$$\text{Var}(X_2) = \Sigma_{22} = 2.$$

Therefore:

$$X_2 \sim N(0, 2).$$

- c. Using property 4 of the multivariate normal (zero covariance implies independence), determine whether X_1 and X_3 are independent. Justify your answer using an entry of Σ . [1 mark]

From the covariance matrix:

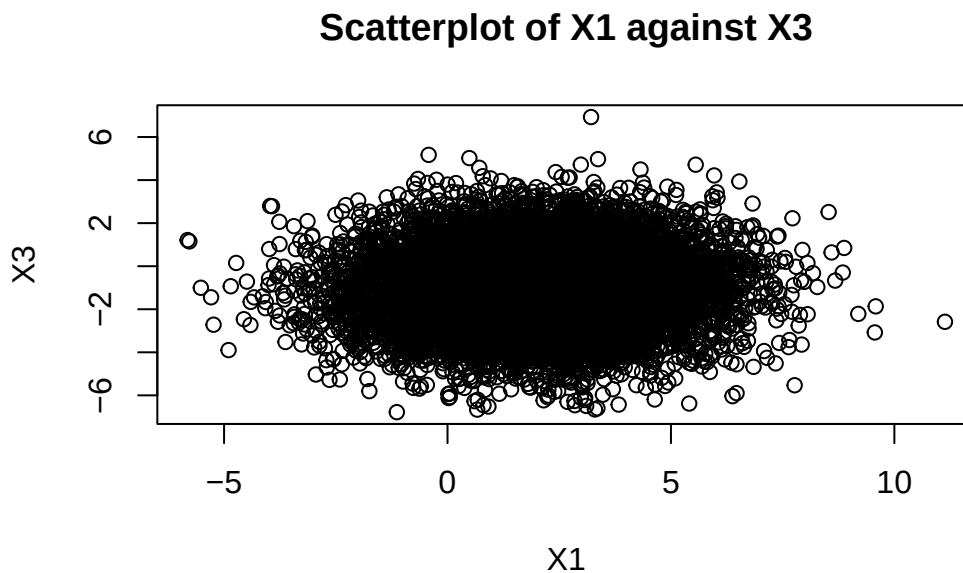
$$\text{Cov}(X_1, X_3) = \Sigma_{13} = 0.$$

Because X is multivariate normal, zero covariance implies independence. Therefore, X_1 and X_3 are independent.

- d. In R, using your simulated draws from Question 10(e), produce a scatterplot of X_1 against X_3 and compute `cor(X1, X3)`. Comment on whether the plot is consistent with your answer to part (c). [3 marks]

```
#Extract simulated variables
X1 <- X_sim[, 1]
X3 <- X_sim[, 3]

#Scatterplot
plot(X1, X3,
     xlab = "X1",
     ylab = "X3",
     main = "Scatterplot of X1 against X3")
```



```
#Sample correlation
cor_X1_X3 <- cor(X1, X3)
cor_X1_X3
```

```
[1] 0.01941785
```

The sample correlation is close to zero, and the scatterplot does not show a clear linear relationship.

⇒ The simulation supports the conclusion that X_1 and X_3 are independent.